

Occurrence, fate, and biodegradation of estrogens in sewage and manure.

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The estrogens estrone (E1), 17alpha-estradiol (E2alpha), 17beta-estradiol (E2beta), and estriol (E3) are natural sex hormones produced by humans and animals. In addition, there are some synthetic estrogens, such as 17alpha-ethinylestradiol (EE2), used for contraception purposes. These compounds are able to produce endocrine disruption in living organisms at nanogram-per-liter levels. In both humans and animals, estrogens are excreted in urine and feces, reaching the natural environment through discharge from sewage treatment plants (STP) and manure disposal units. In STPs, hormone removal depends on the type of treatment process and on different parameters such as the hydraulic and sludge retention times. Thus, hormone elimination rates vary from 0% to 90% in different STPs. Animals are also an important source of estrogens in the environment. Indeed, animals produce high concentrations of hormones which will end up in manure which is typically spread on land. Hence, waste-borne animal hormones may transfer these pollutants to the soil. The purpose of this review is to highlight the significance for both health and the environment of pollution by estrogens and critically review the existing knowledge on their fate and removal in different treatment processes. Relevant information on the microbial degradation of hormones and metabolic pathways is also included.